

DATA202/DATA472 Assignment 3

Due: Thursday, 8 October at 11:59 PM NZST, Worth 8%

- Submit the assignment using the ECS submission system.
- Submit **only** the file `assignment3.Rmd`.
- Make sure the file knits correctly to a .html file. You will need to fix your errors and resubmit if it does not create a .html file.
- **Check your output.** Open the .html file and make sure it contains your answers as you expect to see them.
- The log file contains information about any errors that occurred: check it, too.
- You can resubmit as many times as you like up to the assessment deadline.
- Make sure your name and student ID number are at the top of your file, that your R code is in executable chunks, and any comments are included as text.
- Also please note the **Student Honour Code**, which you agree to when you submit:

When submitting my work I confirm:

- I have completed all steps of the attached assessment on my own,
- I have not used any unauthorised materials while completing this assessment, and
- I have not given anyone else access to my assessment.

Also note:

- Show all R code in your file.
- Show all output where requested to do so.
- Do **NOT** use any libraries other than those you are instructed to use: they may not be on the submission server, and your code will not work.
- Do **NOT** install any packages in your code: when you submit your file it does not have privileges to install anything, and your code will fail.
- Make sure your code works by assuming all external data files it needs or writes to are in the same directory as the .Rmd file. So therefore do not refer to data files using an explicit directory path: refer to them only by their names.

Assignment Questions Q1. (17 Marks)

Write code to simulate a population of individuals and breed it for 20 more generations. Use 50 individuals in the population, 100 genetic markers and 2 copies of each genetic marker in each individual. Keep the data from every generation (i.e. 21 generations including the original one).

Assume that for each child in each new generation, the breeding chooses two different parents at random from the entire previous generation, independently of the parents chosen for any other children.

Each copy of each genetic marker can be variant “0” or variant “1”. The starting frequencies of “0” and “1” variants (alleles) in the population should be 0.6 and 0.4.

Write code to survey 20 individuals from the initial population of 50, and 20 individuals from the last generation of 50. Calculate the relative frequencies of the variant “0” at each marker **in each of those samples**, and plot a histogram of the frequencies for the 20 individuals from the original population, and another histogram for the 20 individuals from the latest generation. Label the histograms appropriately, and comment on them.

Identify all the lines in your code that involve randomness. Also comment on which simulation steps generate a set of simulated values that are independent of each other, and which steps generate values that are not independent of each other. (Note: if you want to add number lines to your code and refer to the line numbers, you can use the `attr.source='.numberLines'` option for your code chunk.)

In the original generation, the only observed frequencies are between 0.4 and 0.8. After 20 generations, the range of observed frequencies of variant “0” over all the markers has expanded to 0 and 1.

The randomness is in lines 14, 21, 55, 67 and 68. Lines 21 and 55 generate sets of independent values. Lines 14, 67 and 68 do not.

Q2. (12 Marks)

Simulate a queueing system in a shop that is open from 9am till 5pm. Assume that customers arrive one after the other such that the distribution of gaps, **in minutes**, between customers who arrive in the morning is `Exponential(1/8)` and the distribution of gaps, **in minutes**, between customers who arrive in the afternoon is `Exponential(1/4)`. (The last gap in the morning is the one until the arrival of the first afternoon customer.)

Simulate 1000 days in the shop and store the overall number of customers who arrive each day. Then plot the histogram of these values and calculate the range and mean over all 1000 replicates. Comment on the results.

Q3. (9 Marks)

Write R code to simulate 10,000 values from a mixture of 5-dimensional repeated-measures multinomial/nominal distributions with 3 categories: - 20% from the distribution with probabilities 10%, 20% and 70% - 50% from the distribution with probabilities 80%, 15% and 5% - 30% from the distribution with probabilities 25%, 60% and 15%

Plot three bar plots, one for each cluster, showing the distributions of the 3 categories. Label the plots appropriately.

Calculate the overall sample mean. What is it an estimate of?

(Assignment total: 38 Marks)